

Closing G-A0-DUI: differentiation under the integral for the slaved variance, and the demotion of H-ANCHOR to a verified anchor fact

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

The $H-A0 \rightarrow H-ANCHOR$ replacement left one textbook residual, G-A0-DUI: the differentiation-under-integral regularity for $M(m)$ underlying Lemma L1 ($M' < 0$). This note writes it out explicitly and closes it, then records the consequence: the $A = 0$ uniqueness becomes an unconditional theorem at the production anchor, so H-ANCHOR demotes from a hypothesis to a verified fact.

2. G-A0-DUI: the explicit argument

With $D(k, m) = m + C(k^2 - q_0^2)^2$ and the radial reduction $M(m) = \frac{1}{2\pi^2} \int_0^\infty \frac{k^2}{D(k, m)} dk$:

Lemma 1 (L1, now unconditional). *For every $m > 0$, M is differentiable with*

$$M'(m) = -\frac{1}{2\pi^2} \int_0^\infty \frac{k^2}{D(k, m)^2} dk < 0.$$

Proof. Fix $m > 0$ and set $m_0 = m/2$; for $|h| < m/2$ every intermediate value $\tilde{m} \in [m/2, 3m/2]$ satisfies $\tilde{m} \geq m_0$, so the SAME dominator applies on BOTH sides ($h \gtrless 0$), giving two-sided differentiability at m . For $\tilde{m} \geq m_0$, $\partial_{\tilde{m}} D^{-1} = -D^{-2}$ satisfies the pointwise bound

$$\left| \partial_{\tilde{m}} \frac{1}{D(k, \tilde{m})} \right| = \frac{1}{D(k, \tilde{m})^2} \leq \frac{1}{D(k, m_0)^2} =: \frac{1}{[m_0 + C(k^2 - q_0^2)^2]^2},$$

and the dominating function $k^2/D(k, m_0)^2$ is integrable on $(0, \infty)$ across all three regions: near $k = 0$ the denominator is bounded below by a positive constant so the integrand is $O(k^2)$; near the shell $k = q_0$ it is bounded by k^2/m_0^2 (where $D \rightarrow m_0$); and as $k \rightarrow \infty$ it decays as $k^2(Ck^4)^{-2} = C^{-2}k^{-6}$. By the Leibniz rule under dominated convergence, differentiation and integration commute, giving the stated M' ; the integrand k^2/D^2 is strictly positive, so $M'(m) < 0$ strictly. $\square$

Machine confirmation (independent radial quadrature): $M' = -\frac{1}{2\pi^2} \int k^2/D^2$ matches the finite-difference M' to 0.04–0.6% at $m = 0.1, m^*, 1.0$; the dominating-function integral is finite (2.36 at $m_0 = 0.05$); the pointwise domination $D(m^*)^{-2} \leq D(m_0)^{-2}$ holds with max ratio 1.000. G-A0-DUI is CLOSED.

3. Consequence: A-zero uniqueness is unconditional at the anchor

With L1 now rigorous and L2 ($g' \leq -1$ on $M \geq M_c$, closed-form since $u_{\text{eff}}(M_c) = u + 10vM_c = 0$ exactly) and L3 ($g \leq m_w - m$ on $M < M_c$, closed-form since $3uM \leq 0$ and $15vM^2 \leq 15vM_c^2$) already closed-form, the sign-decomposition theorem has NO remaining analytic gap. Its only inputs are the two verified closed-form anchor facts

$$m^* > m_w (\times 7.76), \quad M_R > M_c (\times 4.12),$$

machine-verified in G-A0-VER (14/14). Hence the $A = 0$ uniqueness and zero-at-gap structure (former H-A0 (U),(Z)) is an UNCONDITIONAL theorem AT THE PRODUCTION ANCHOR $\mu^2 = 0.005$ ONLY (the μ^2 -neighbourhood is ROBUSTNESS-MU2, OPEN). This is the slaved isotropic $A = 0$ section, NOT a global H-LAYER closure: H-LAYER remains a named hypothesis of B2/B1, and the $A \neq 0$ branch / anisotropic perturbations / off-diagonal covariance are untouched. Permitted: “ $A = 0$ uniqueness is unconditional at the anchor.” Forbidden: “H-LAYER is now unconditional.”

4. H-ANCHOR demotion

A hypothesis is an assumption that could fail; $m^* > m_w$ is an established closed-form inequality on the anchor constants ($m^* = r + 3uM_R + 15vM_R^2 = r_R$, $m_w = r + 15vM_c^2$), verified with a $\times 7.76$ margin. It is therefore a VERIFIED DEPENDENCY, not a hypothesis. H-ANCHOR is demoted accordingly and REMOVED from the hypothesis sets of B2 and B1:

$$\begin{aligned} \text{B2} &: \{\text{H-LAYER}, \text{H-ANCHOR}\} \rightarrow \{\text{H-LAYER}\}, \\ \text{B1} &: \{\dots, \text{H-ANCHOR}, \dots\} \rightarrow \{\text{H-LAYER}, \text{H-ADM-COH}, \text{SC-SCOPE}\}. \end{aligned}$$

The dependency is RETAINED, not erased: H-ANCHOR leaves the hypothesis set but remains a verified anchor dependency — $m^* > m_w$ and $M_R > M_c$ under A1-KERNEL-CONV at $\mu^2 = 0.005$ — recorded as a verified-fact row in GATES.md and in the B1/B2 card scope. Demoting the hypothesis does NOT hide the anchor dependence; the curve-shape quadrature systematic exits the uniqueness proof, but the anchor constants remain verified dependencies (this is not “all numerical dependence removed”). The OFF-ANCHOR extension (the μ^2 -neighbourhood) is the pre-existing ROBUSTNESS-MU2 gate, unchanged: this demotion concerns only the production anchor, where the whole Sector-B analysis is pinned.

5. Numerical verification

Script [codes/vacuum/ha0_sign_decomposition.py](#) (v1.1.0; 23/23 self-test asserts PASS, the 9 new G-A0-DUI asserts included; artefact [claims/B2-PROPA-HLAYER/runs/260606-ha0-sign-decomposition/result.json](#)).

Quantity	Value	Check
$M' = -\frac{1}{2\pi^2} \int k^2/D^2$ vs FD	agree 0.04–0.6%	formula confirmed; both < 0
dominating-function integral	$2.36 < \infty$ at $m_0 = 0.05$	k^{-6} tail integrable
pointwise domination	max ratio $1.000 \leq 1$	$D(m^*)^{-2} \leq D(m_0)^{-2}$
$M'(m^*)$ sign	< 0 strictly	L1 rigorous, G-A0-DUI CLOSED
$u_{\text{eff}}(M_c)$	$= 0$ exactly	L2 boundary closed-form

Quantitative sanity check (mandatory): the two independent quadratures of $\int k^2/D^2$ (the closed-form M' and the finite-difference of M) agree to $\leq 0.6\%$ – discretisation level, confirming the analytic identity, not asserting an exact numerical match (the rigour is the dominated-convergence argument, not the quadrature).

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3.5)

- (α) “Is removing H-ANCHOR from the hypotheses just relabelling to inflate the result?” DISMISSED: $m^* > m_w$ is a proven closed-form inequality (G-A0-VER, $\times 7.76$), not an assumption; a proven fact is a dependency, not a hypothesis. The theorem stated is “ $A = 0$ uniqueness at $\mu^2 = 0.005$, unconditional” – exactly what is established; no scope is inflated.
- (β) “Is the DUI really closed, or is the 0.6% quadrature gap a hidden problem?” DISMISSED: the rigour is the dominated-convergence argument (stated dominating function, integrable, pointwise domination) – ANALYTIC, exact. The 0.6% is the agreement of two independent numerical quadratures, a CONFIRMATION of the formula, not a load-bearing bound.
- (γ) “Does dropping H-ANCHOR weaken the honest tracking of the anchor dependence?” VALID-with-mitigation: to keep it visible, the GATES.md H-ANCHOR row is retained as a VERIFIED-FACT entry (not deleted), and the cards’ scope states the inequality; only the hypothesis CLASSIFICATION changes. The off-anchor question stays in ROBUSTNESS-MU2.
- (δ) “Tier monotonicity after B2 drops to one hypothesis?” DISMISSED: B2 stays T6 with the single named hypothesis H-LAYER; the linter’s T6-with-sub-T6-dep rule needs ≥ 1 named hypothesis, satisfied. B1 keeps three named hypotheses.

No objection upheld as a blocker.

7. Result footer

Result ID:	B2-PROPA-HLAYER / G-A0-DUI closure + H-ANCHOR demotion
Precise statement:	L1 ($M' < 0$) proven by dominated convergence with the dominating function $k^2/[m_0 + C(k^2 - q_0^2)^2]^2$ (integrable; pointwise domination); G-A0-DUI CLOSED. With L2/L3 closed-form, $A=0$ uniqueness is unconditional at $\mu^2=0.005$, resting only on the verified facts $m^* > m_w$ ($\times 7.76$), $M_R > M_c$ ($\times 4.12$). H-ANCHOR demoted hypothesis $\rightarrow$ verified fact; removed from B2 and B1 hypothesis sets.
Scope:	Production anchor $\mu^2 = 0.005$. Off-anchor = ROBUSTNESS-MU2 (unchanged).
Dependencies:	A1-KERNEL-CONV (anchor constants); the gap equation.
Evidence grade:	ANALYTIC (dominated convergence) + EXECUTED (23/23)
Reproduction command:	python codes/vacuum/ha0_sign_decomposition.py
Expected output:	claims 23/23 PASS; M' formula vs FD $< 0.6\%$; dominating integral finite; result.json under claims/B2-PROPA-HLAYER/runs/260606-ha0-sign-decomposition/
Falsification gate:	A non-integrable dominating function (would break

the DUI); $M' \geq 0$ somewhere (refutes L1); $m^* \leq m_w$ at the anchor (refutes the verified fact).

Tier before / after: B2 T6 (hyps {H-LAYER, H-ANCHOR} $\rightarrow$ {H-LAYER});
 B1 T6 (hyps drop H-ANCHOR $\rightarrow$ {H-LAYER, H-ADM-COH, SC-SCOPE}). No tier-number change; both strengthen.

No-overclaim statement: The result is unconditional ONLY at the production anchor; the μ^2 -neighbourhood is ROBUSTNESS-MU2, open and unchanged. G-A0-DUI is closed analytically; the numerics confirm, they do not carry the proof.
 No tier-number change on B1/B2.

Next required action: ROBUSTNESS-MU2 (off-anchor neighbourhood) is now the sole residual of the former H-A0; SC-SCOPE all-orders and ESTIMATOR-UPGRADE remain for B1.